Uncertainty from b quark masses in Z pT spectrum

Status summary

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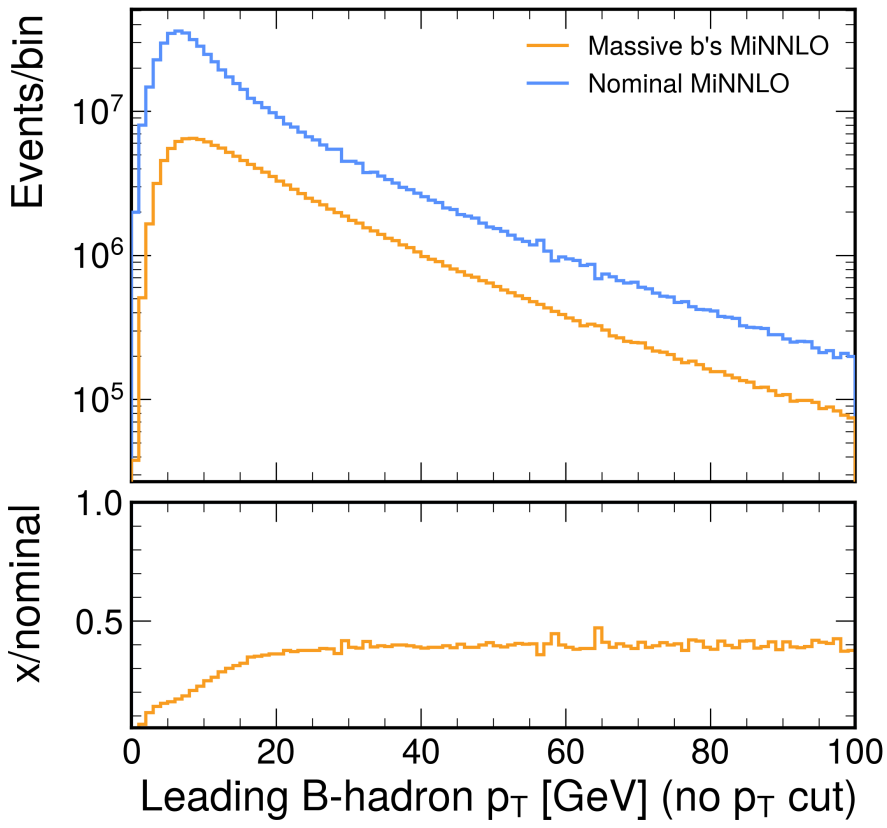
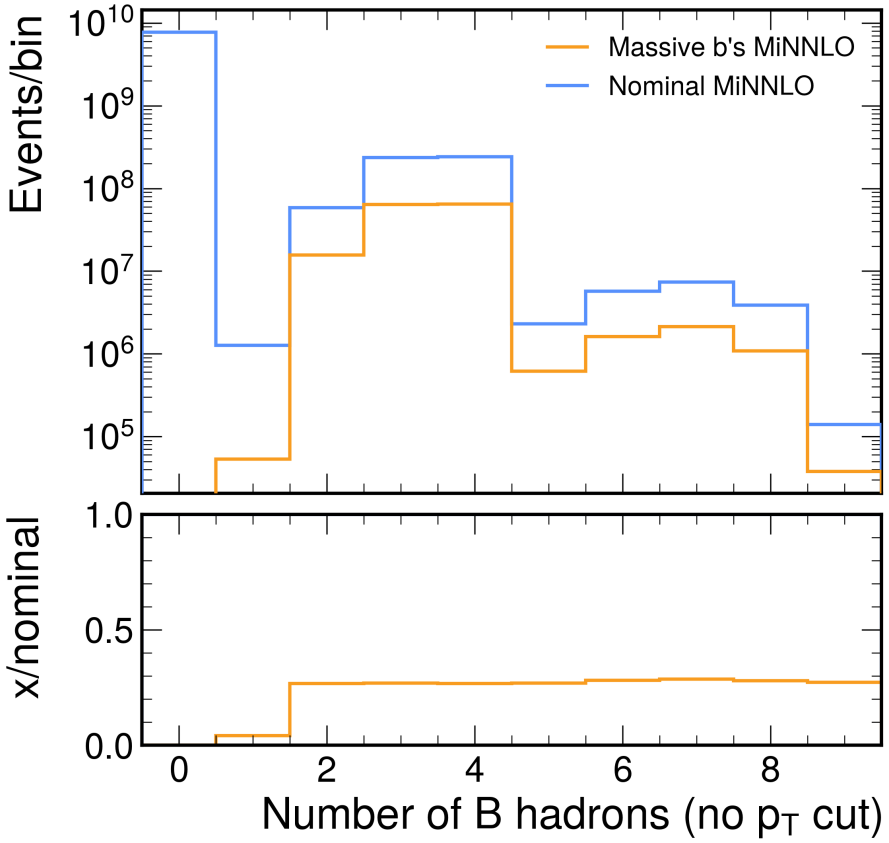
Overview: Samples, Goal, Methodology

- Goal: build a b-mass uncertainty to apply to our simulation, since our nominal MiNNLO treats b's as massless.
  - N.B.: we are already accounting for heavy quark mass effects in PDFs.
- Method:
  - select events affected by heavy b quark mass.
  - Build a corrected distribution as massless-b MiNNLO (unselected) + massive-b MiNNLO (selected).
  - The difference between the corrected distribution and the nominal MiNNLO is taken as an uncertainty.
- We attempted to use the B hadrons to define the set of events that should be swapped, but it doesn't seem to work nicely. We looked at jets and LHE quarks as well, see backup.
- Samples:
  - nominal MiNNLO: massless b, 5FS
  - dedicated heavy b quark $Zb\bar{b}$ MiNNLO: massive b, 4FS. (From arXiv2404.08598: $\sigma = 25.28$pb from Table 1, but unclear if that is inclusive $\ell = e, \mu$ or for each lepton flavor.)

B-Hadron Diagnostics

- B hadrons are identified from GenPart using PDG-ID hadron-content logic (contains bottom flavor) with GenPart status 1 or 2.
  - Exact implementation is shown in backup.

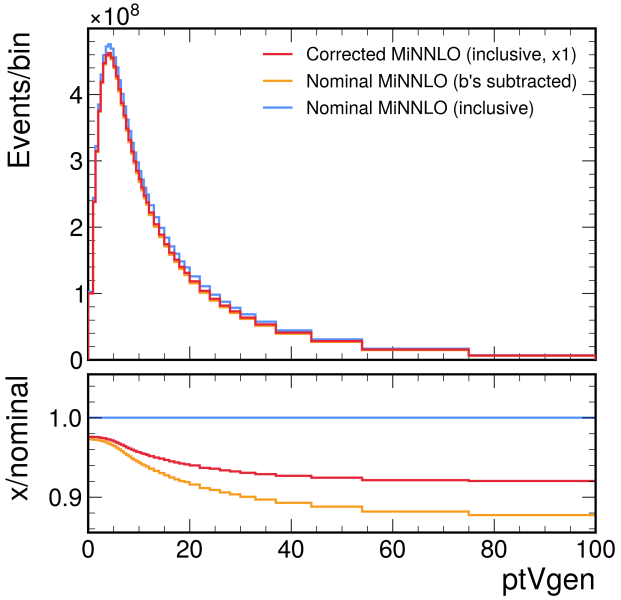




- Large, unexplained normalization difference, on top of what may be a reasonable shape difference at low p_T.

Swap Result and Definition

- Swap definition: select events with at least one such B hadron (no p_T selection); corrected distribution is built as nominal MiNNLO (unselected) + massive-b MiNNLO (selected).

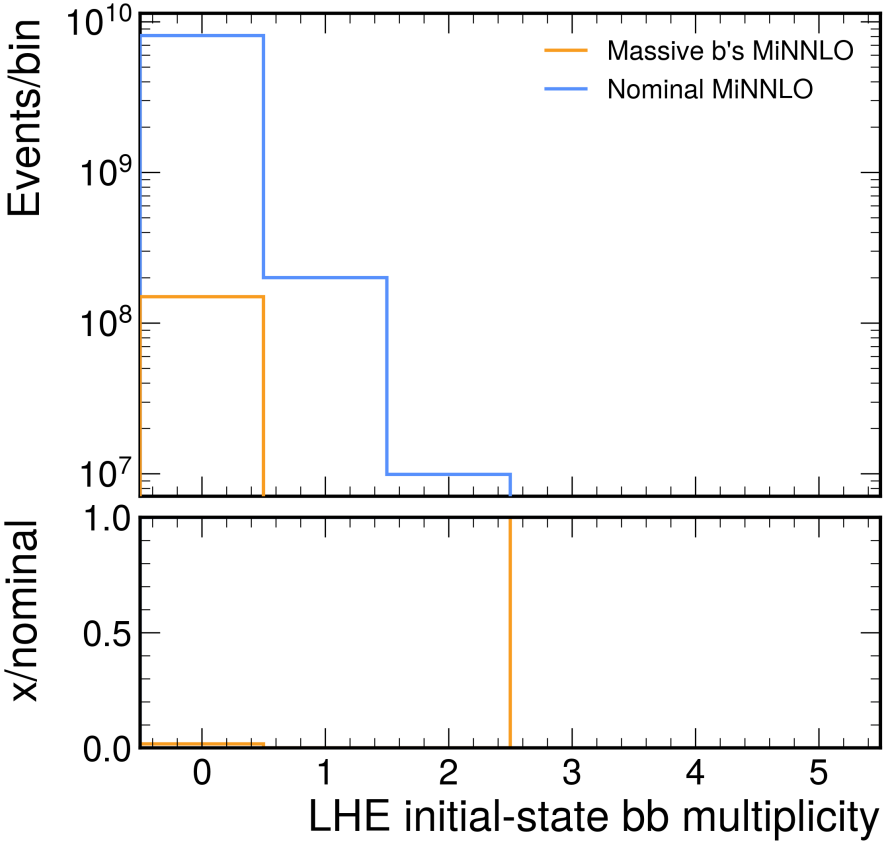


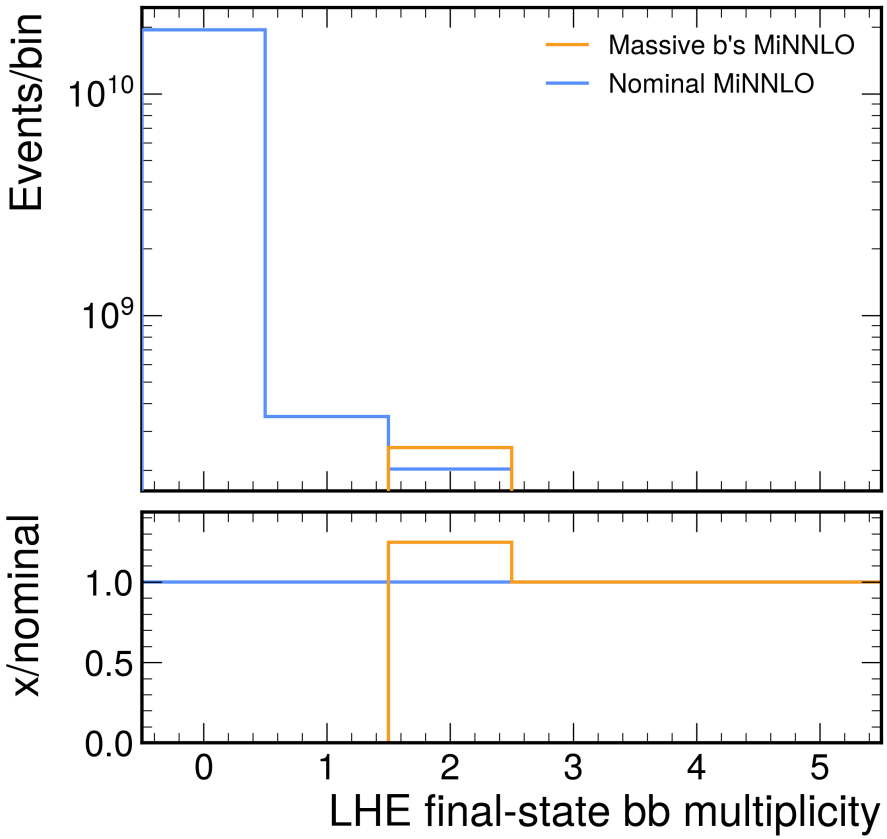
- Result: swapping gives a sizable normalization shift (not shape-only), so it is too aggressive as-is.
- Questions: is this due to correct normalization difference between samples? Are we selecting more events from the nominal MiNNLO than the ones we are putting back in from the massive-b MiNNLO? Or is it something else?

Backup

Backup: LHE Multiplicity

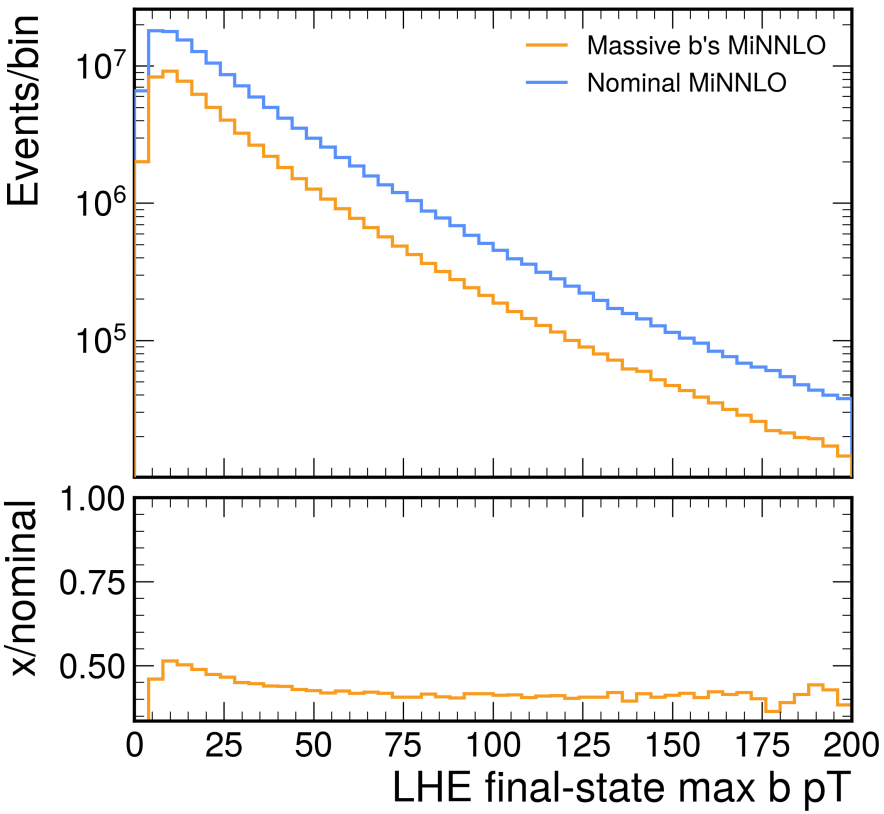
- LHE *b*-quark observables built with $|\mathrm{pdgId}| = 5$ and final-state status.

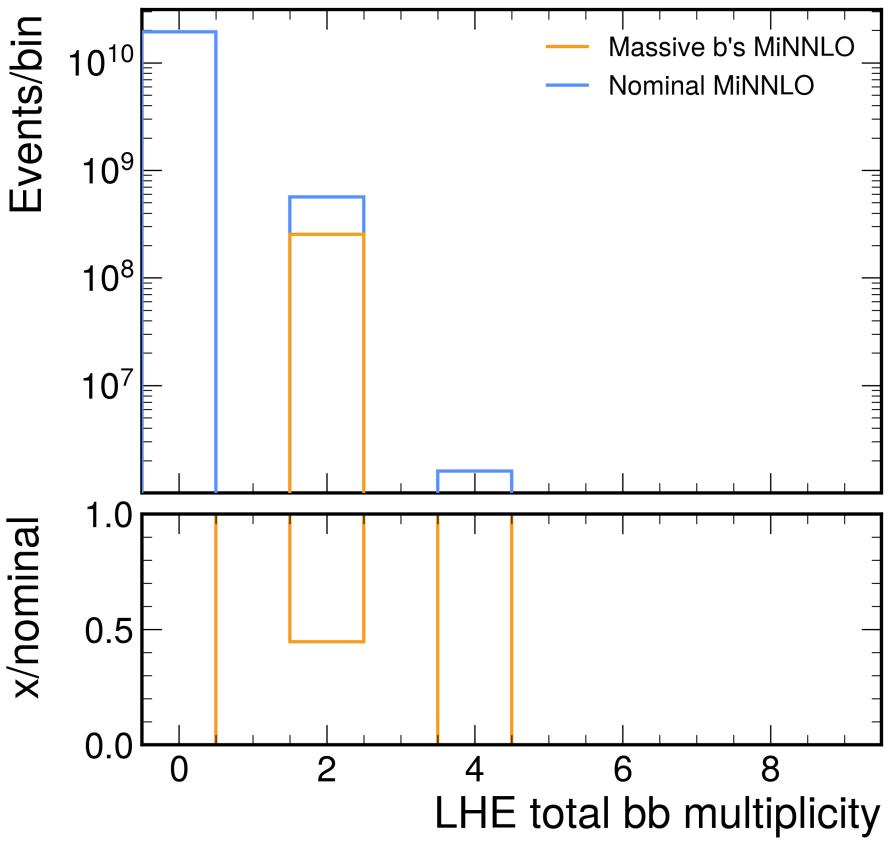




Backup: LHE Leading-b p_T and Total Multiplicity

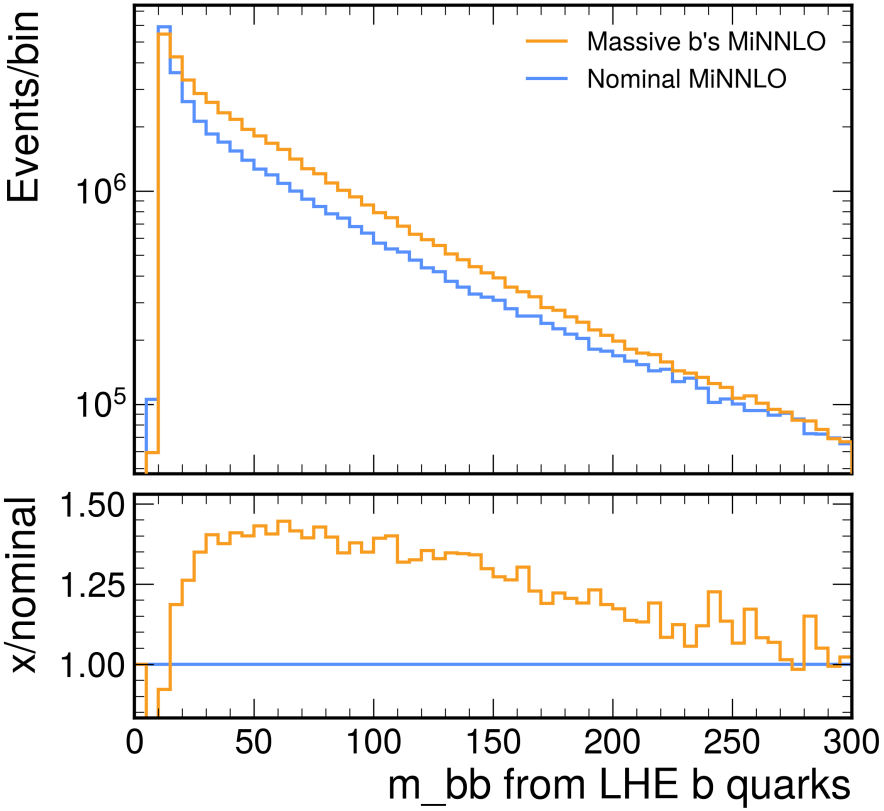
- LHE b-quark observables built with $|\mathrm{pdgId}| = 5$ and final-state status.

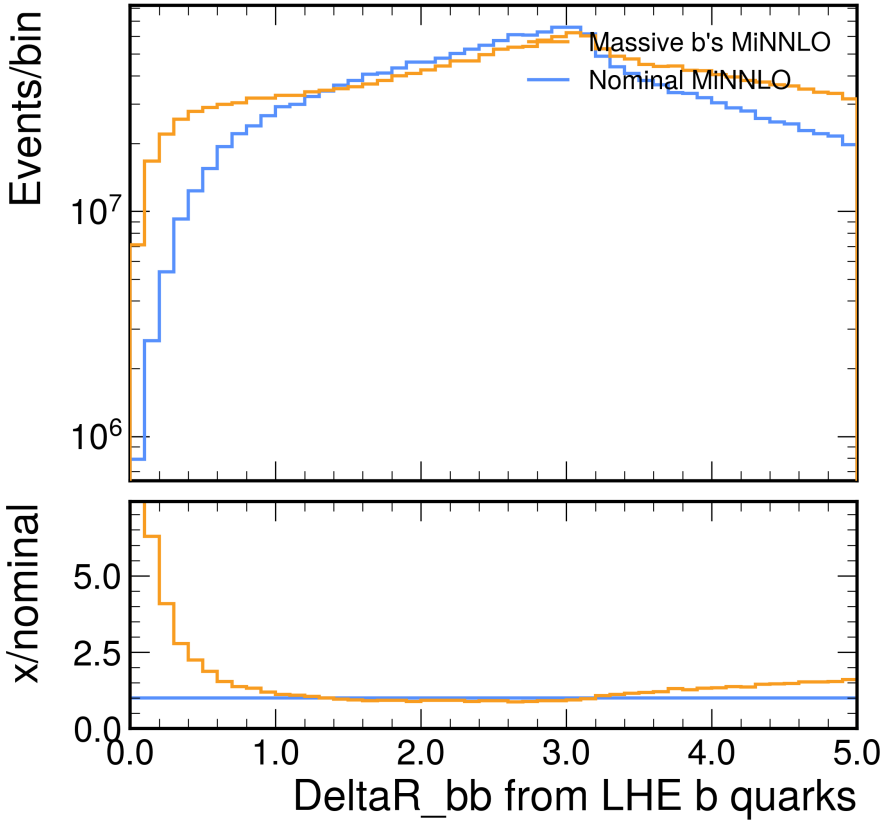




Backup: LHE $b\bar{b}$ Kinematics

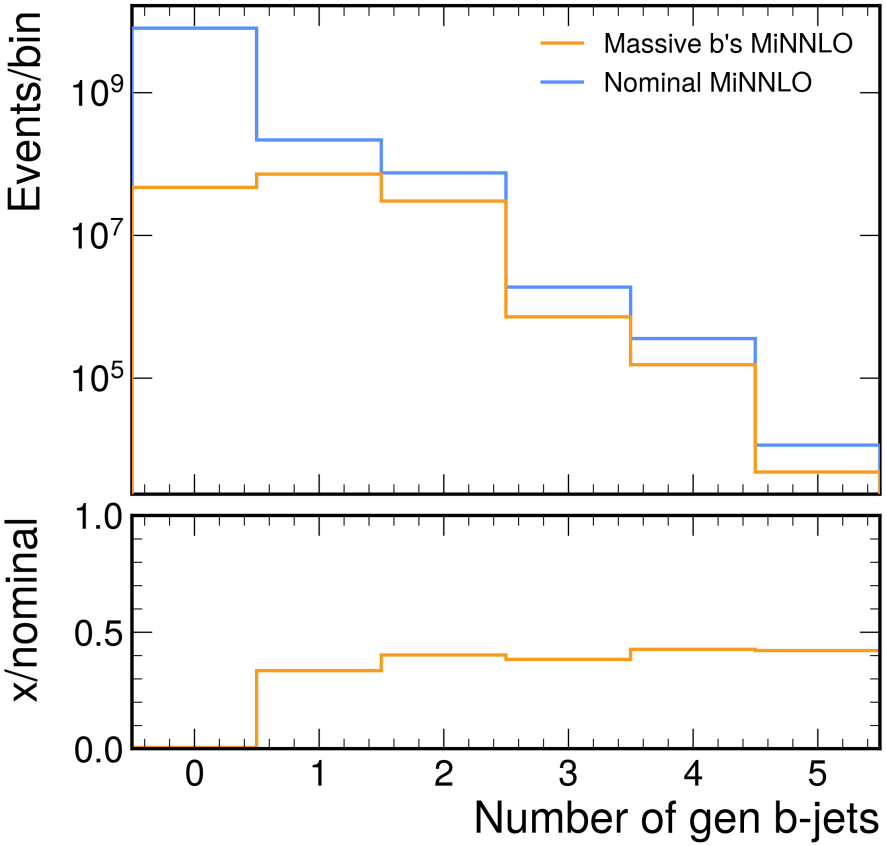
- LHE b-quark observables built with $|\mathrm{pdgId}| = 5$ and final-state status.

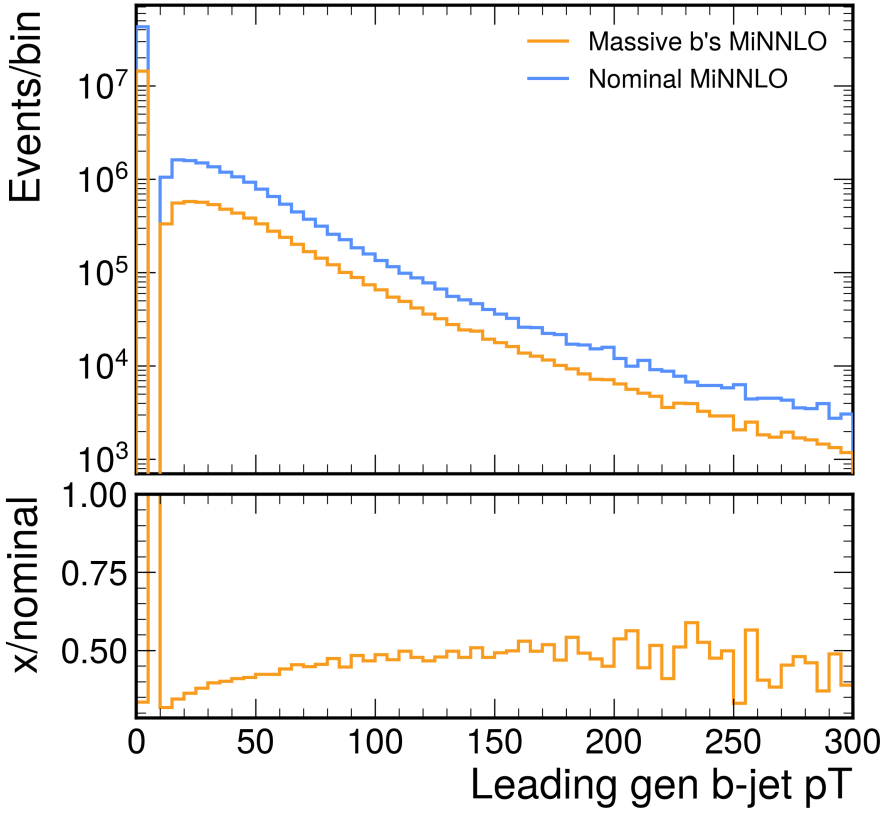




Backup: GenJet b-jet Observables

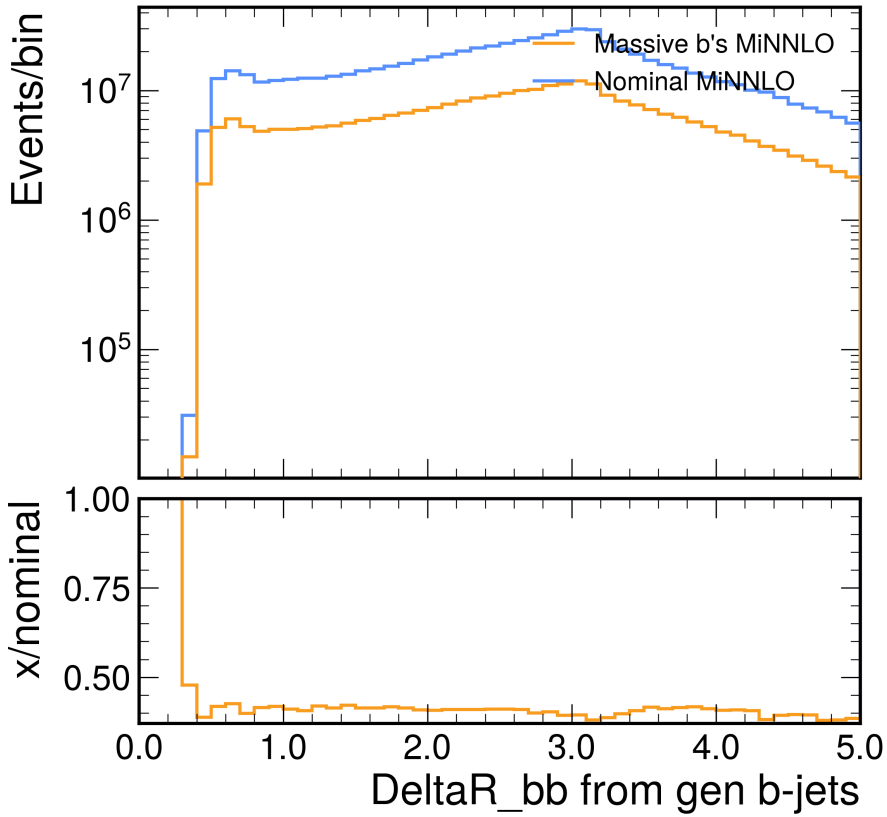
- Gen b-jet observables here use hadronFlavour $= 5$ with no kinematic requirement ($p_T > 0$, $|\eta| < 10$).

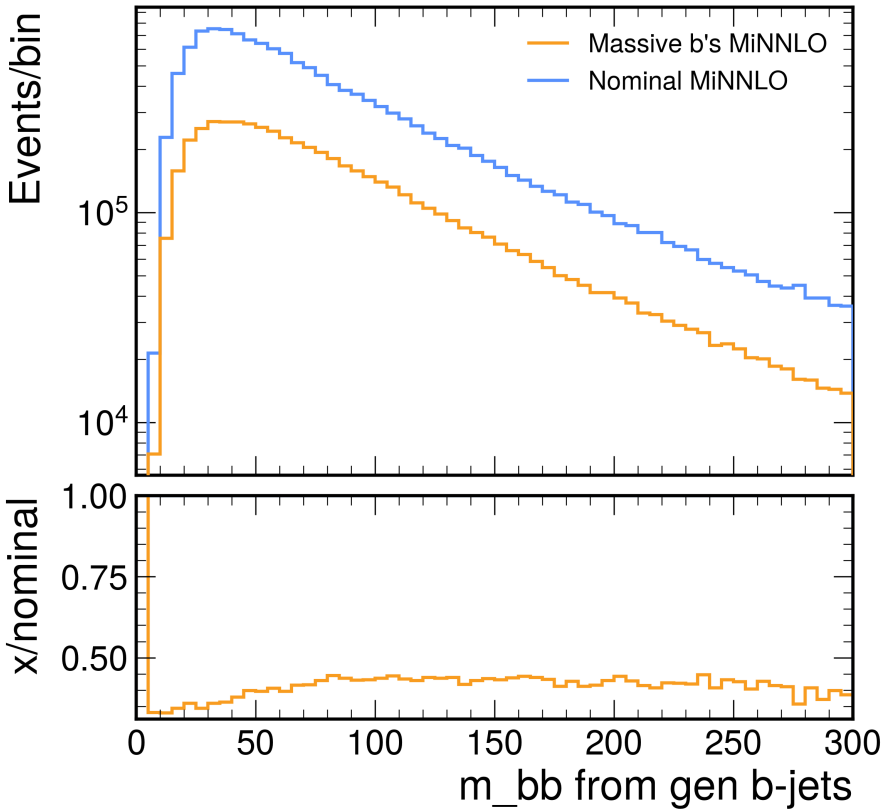




Backup: GenJet ΔR_{bb} and m_{bb}

- Jet-level pair observables from gen b-jets, with the same no-cut jet definition (hadronFlavour $= 5$, $p_T > 0$, $|\eta| < 10$).

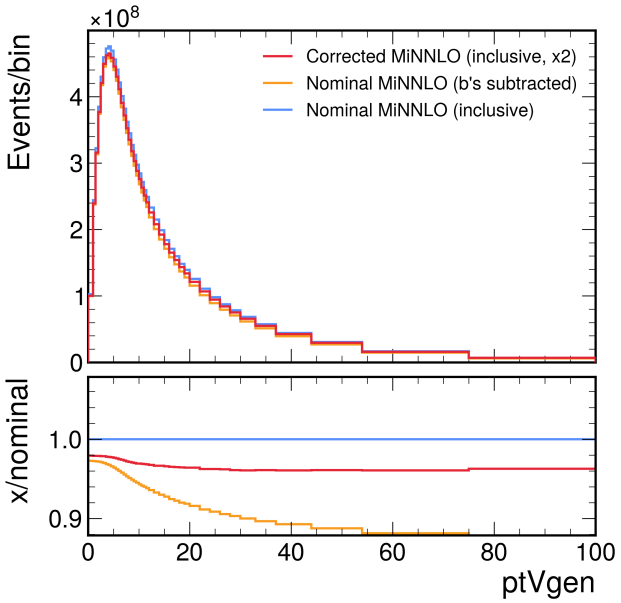




Backup: Exact B-hadron Selection Code

Implementation

```
inline bool isBHadron(int pdgId)
{
    int64_t id = std::llabs(static_cast<int64_t>(pdgId));
    // Non-hadrons / fundamentals
    if (id < 100) return false;
    // Reject nuclei/ions.
    if (id >= 1000000000LL) return false;
    int nq1 = static_cast<int>((id / 1000) % 10);
    int nq2 = static_cast<int>((id / 100)  % 10);
    int nq3 = static_cast<int>((id / 10)   % 10);
    // Exclude diquarks.
    if (nq3 == 0) return false;
    return (nq1 == 5) || (nq2 == 5) || (nq3 == 5);
}
```

Backup: Unnormalized Swap with 4FS Scaled by 2

- Same unnormalized swap definition as main slide, but with an extra factor-2 scaling applied to the selected 4FS component before replacement.